

SUMMER 2022

Liquid Propulsion

GBM 1

2022/07/02

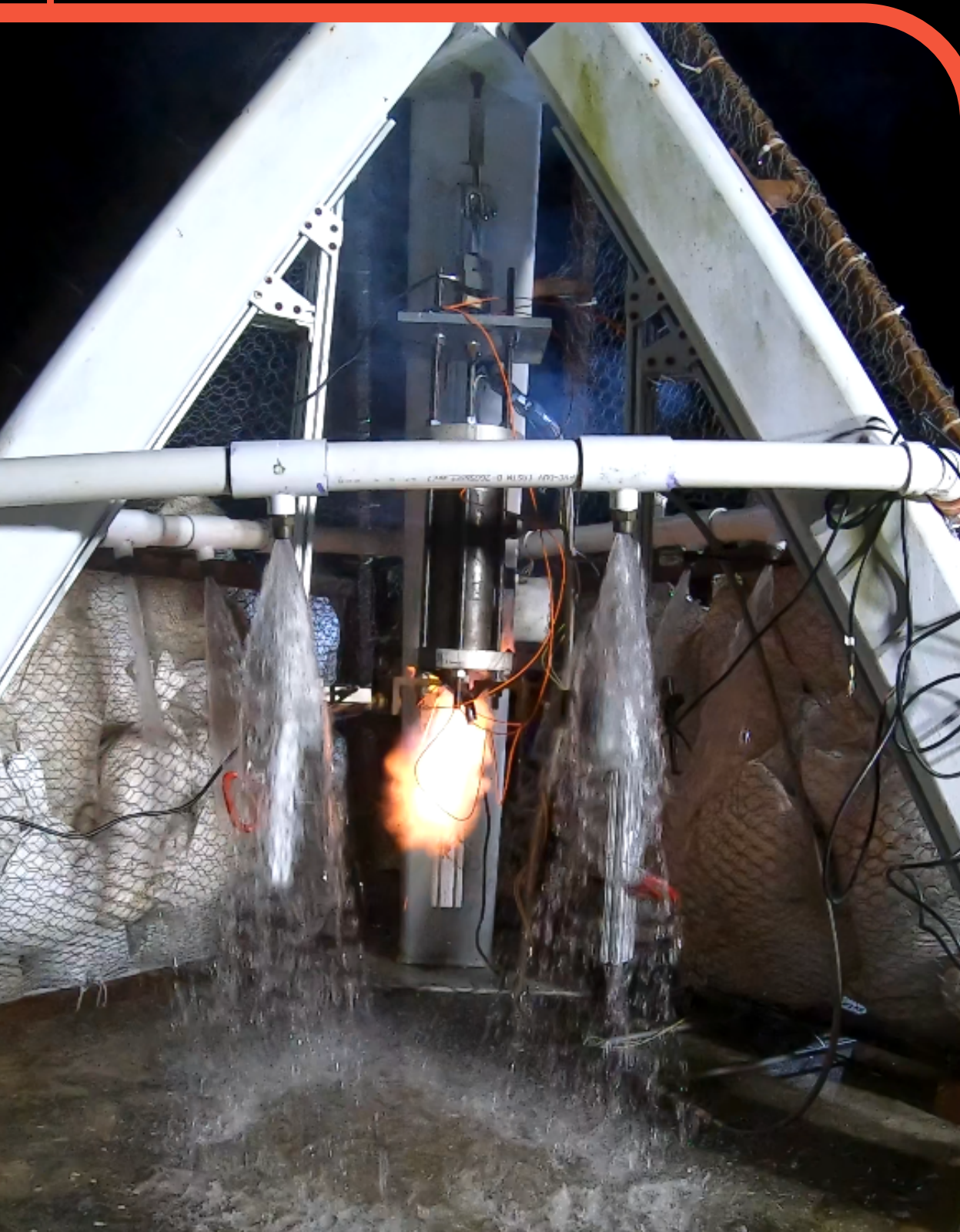
@UFLIQUIDPROPULSION

The Liquid Propulsion Development Team was founded as an entirely undergraduate design team that's focus is to design, manufacture, and fly liquid powered rockets.



Fundamentals

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UF LIQUID PROPULSION

01. DESIGN

We completely design from the ground up all of our testing infrastructure, engines, and future flight vehicles. All of which comes from the knowledge LPDT members learn themselves.

02. BUILD

Everything that we design is eventually build, tested, and iterated for reliability and safety of our members.

03. TEST

Each system that we create is eventually tested with the end goal of achieving a hot fire and then flight.

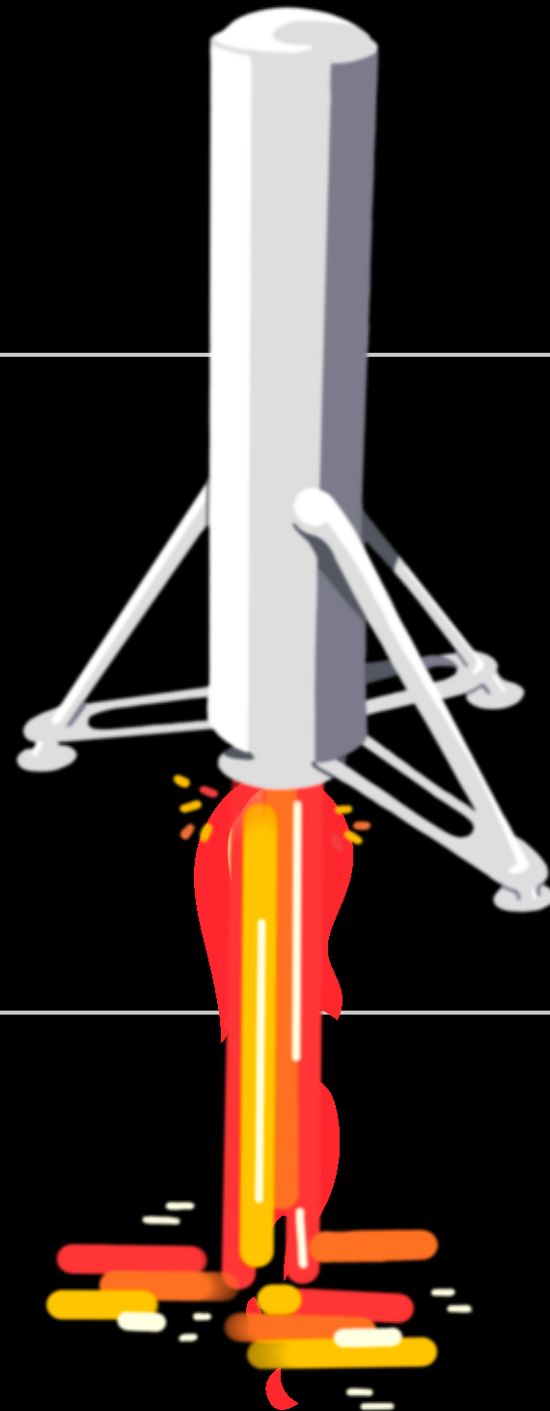
LPDT Update

These were the main projects that have been completed during the spring and summer semester in preparation for the coming year.

Test Stand
Trailer
Insurance Plan
Test Site
Sparrow V1 Design
Electronics System

Completed for Fall
2022

Where are we going?



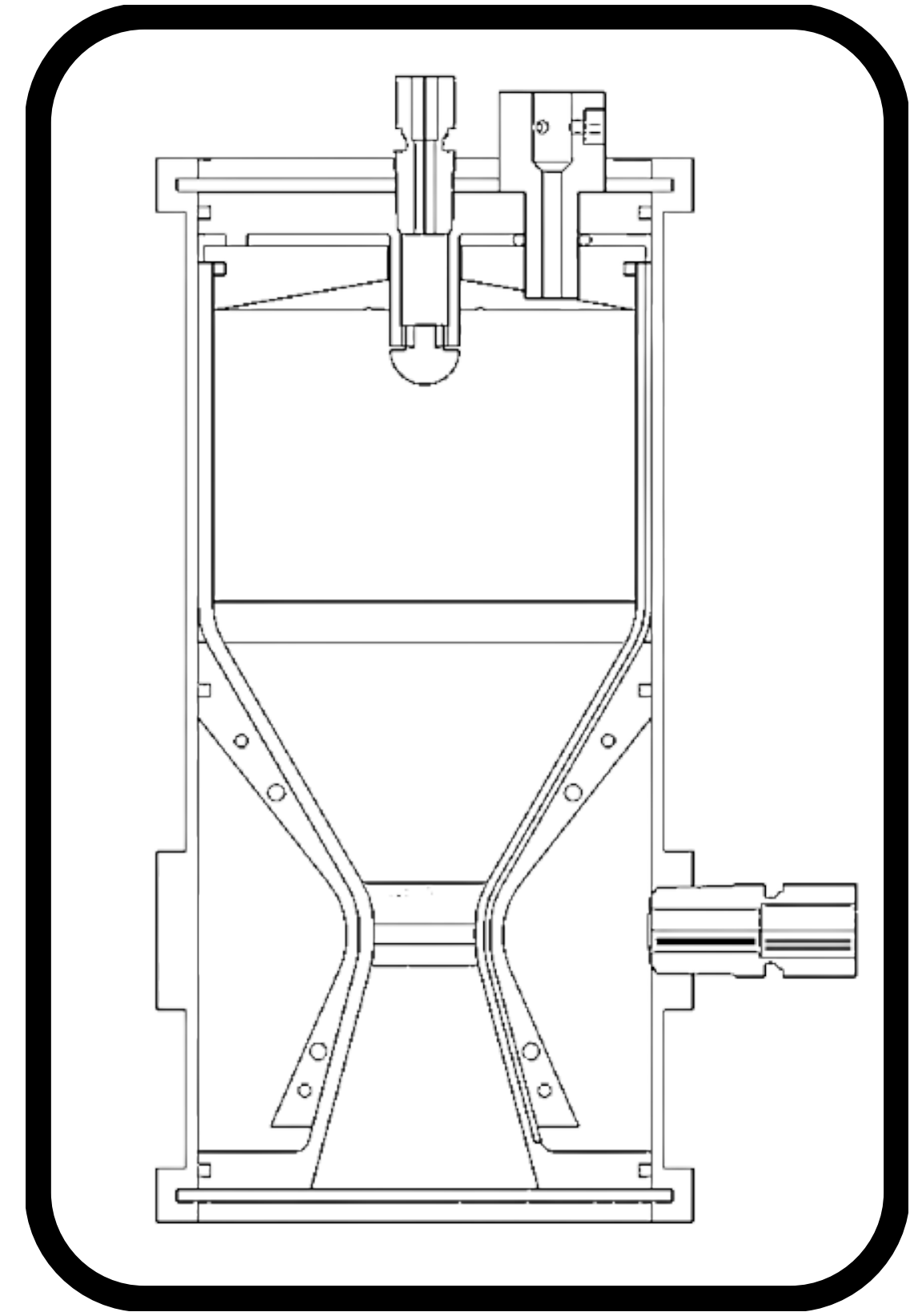
For the first time in LPDT's history we will be competing in a challenge specifically designed for liquid propulsion teams.

As most teams accross the country set their sites on reaching space first. We are aiming to be the first team to fly and land a rocket.

Engine Design

The Engine Design sub team's focus is to mathematically design our rocket engine and create the overall architecture of how the engine will look.

-Jack ten Hulzen



Manufacturing

The Manufacturing sub-team deals with the building of any designs created by the various sub-teams.

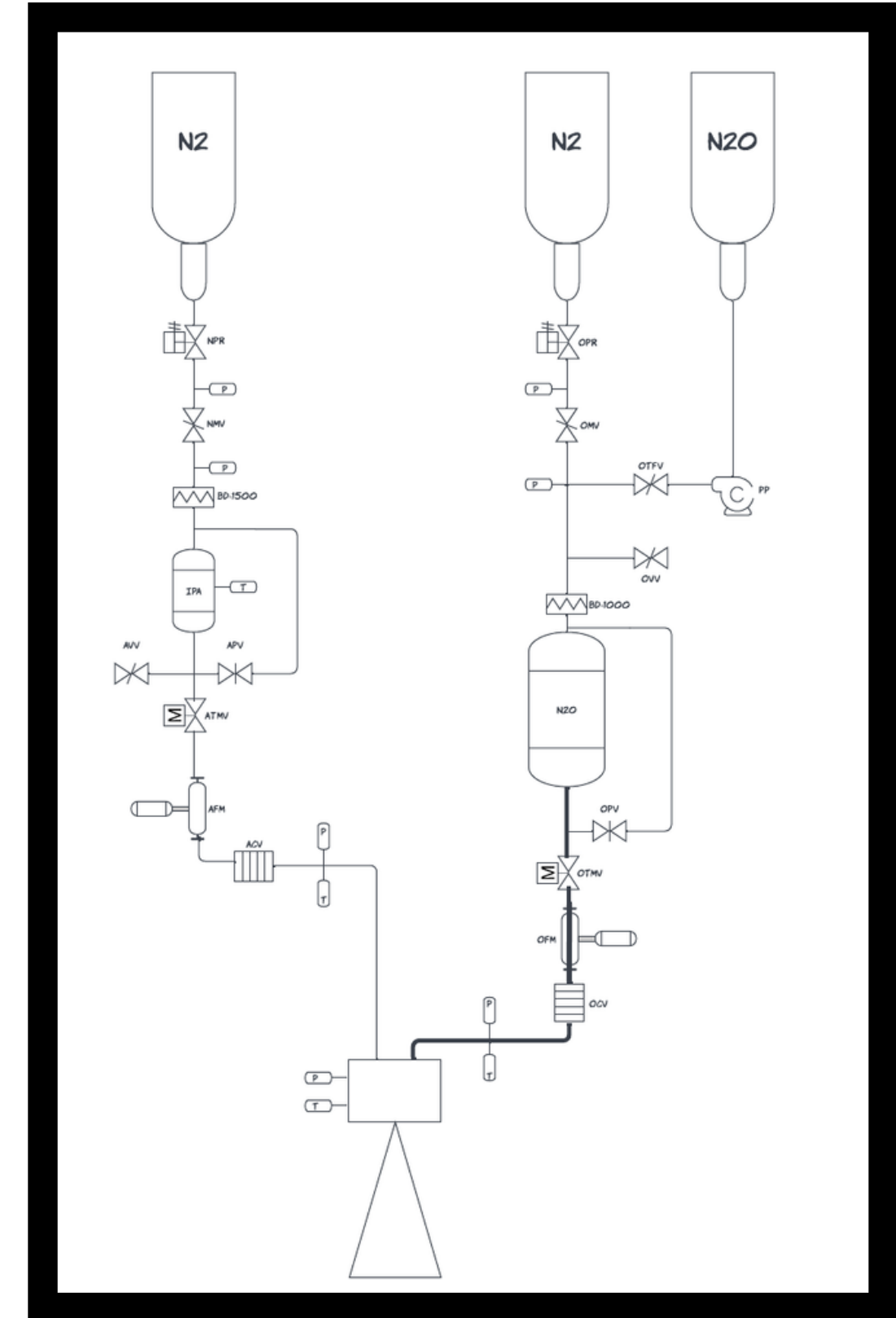
-Kevin Nguyen



Propellant Delivery

The Propellant Delivery sub-team is in charge of designing all of the feed systems that supply the engine with fuel and oxidizer.

-Joey Johnson



Electronics

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The electronics team develops the necessary equipment to run tests. This includes all sensors and associated devices as well as the software to control the engine and visualize data in real-time. This is a great team for not just engineers but computer science and EE majors as well.

-Chase MacDonnell

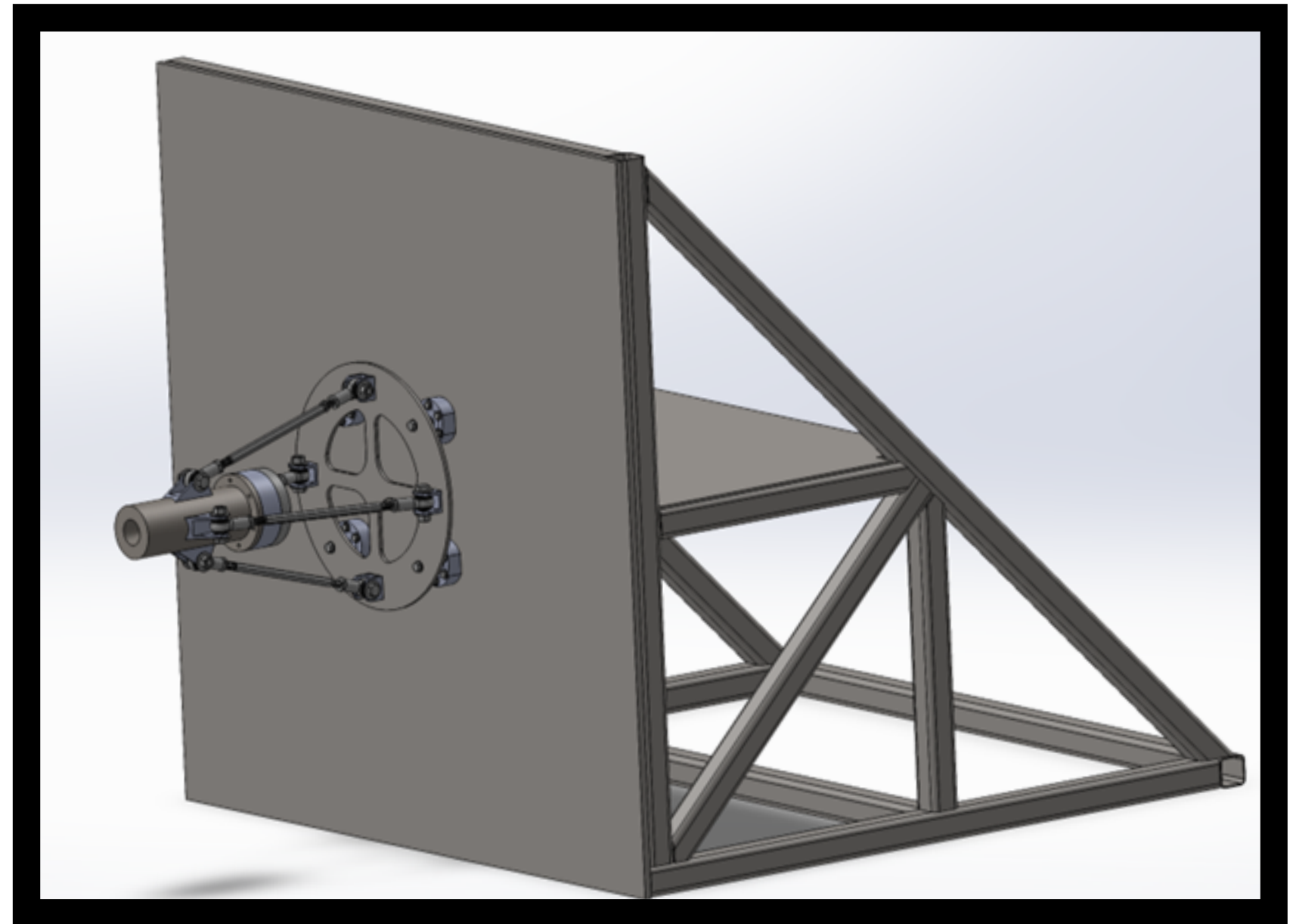


Testing

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The Testing team handles all of the design when it comes to developing the infrastructure needed to safely test any part of hopper. All of the tests conducted by LPDT also have standard operating procedures. It is our job to create these procedures on how to perform various tests.

-Alec Sutton



Safety

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The Safety team is tasked with the evaluation, analysis, and management of any and all potential failures and hazards. Through core functions such as fault management, and systems safety theory, the Safety team ensures the protection of personnel, the safeguarding of hardware, and the overall probability of success for LPDT.

-Lucca Fogaca



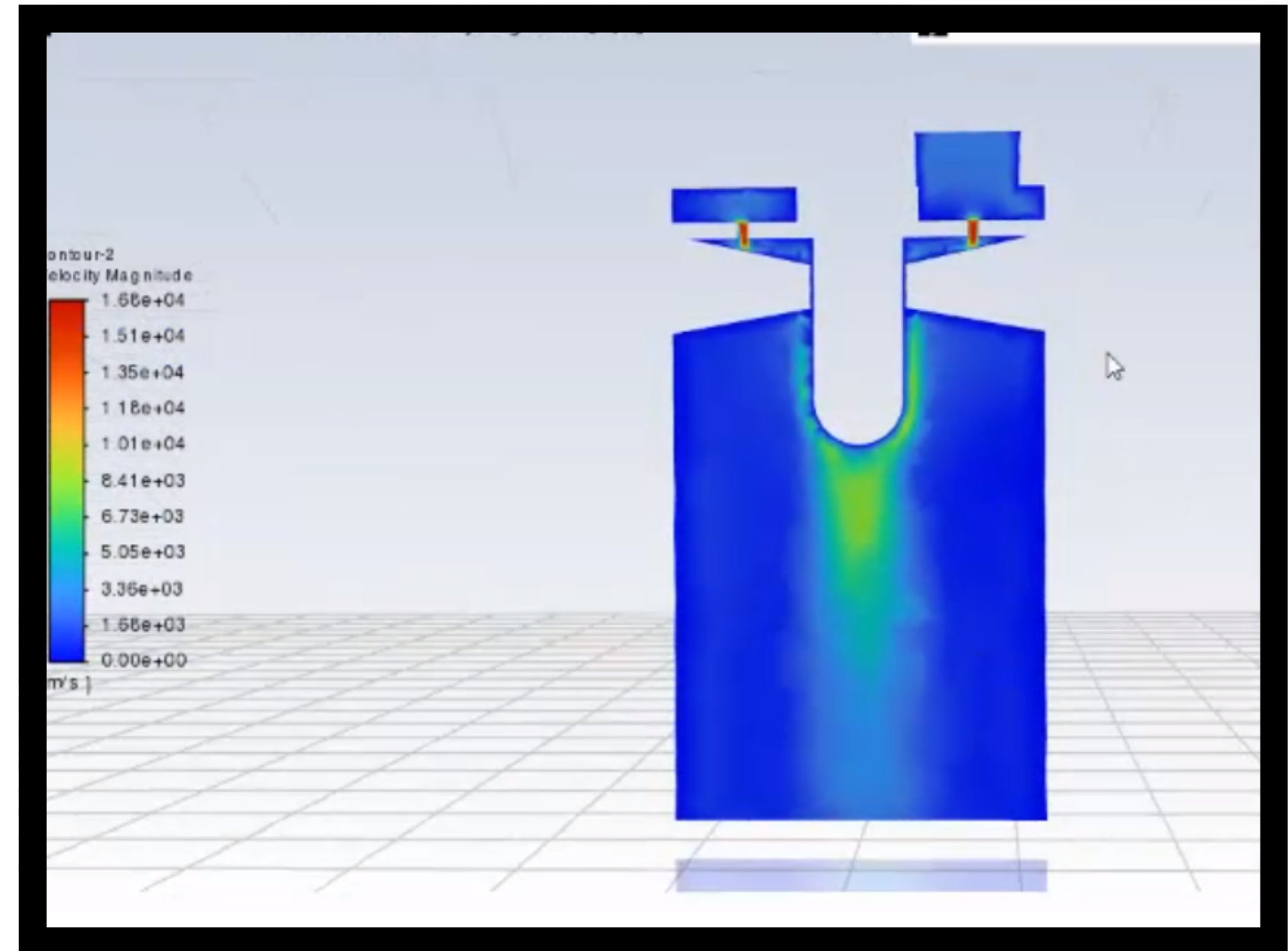
Simulations

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The sims team uses CFD software such as ANSYS Fluent and SOLIDWORKS Simulation to validate engine designs.

We aim to model properties such as pressure drop and spray interaction and compare these with test data to improve engine performance.

-Esha Shah



Business

The Business team focuses on developing company relationships to secure future funding along with internship opportunities. Just this semester we raised \$10,000 dollars. If you are interested in graphic design, website building, or research proposals we act out all the marketing plans for LPDT.

-Nathan O'Brien



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